Liu Jiacheng

+86 199 0560 3151 | jiacheng.liu2004@outlook.com

EDUCATION

The University of Manchester, United Kingdom Sep 2026 – Sep 2027 (Expected)
MSc in Robotics

Hefei University of Technology (Project 211 University), China Sep 2022 – Jun 2026
B.Eng. in Industrial Engineering

- Arithmetic Mean Score: 83%
- Relevant Coursework: C/C++ Programming, MATLAB, Engineering Mechanics, Mechanical Manufacturing, Automotive Construction, Manufacturing System Simulation, Intelligent Production and Process, AutoCAD.

RESEARCH & PROJECT EXPERIENCE

Tire Factory Logistics Simulation & Curing-Section Scheduling Optimization Jul 2026

- Conducted facility-layout and material-flow analysis using SLP methodology to diagnose logistics inefficiencies across tire-factory storage, handling and curing operations.
- Developed a FlexSim discrete-event simulation model for green-tire transportation, simulating flow merging, robotic-arm loading, HBS inbound/outbound handling and curing-machine feeding processes.
- Built a Python genetic-algorithm scheduling model for 10 curing presses and 147 production orders, considering operational constraints to improve order allocation and curing-section scheduling efficiency.

Graduation Design: Research on Algorithms and Systems for Human Fall Detection Based on Skeletal Keypoints Jun 2026

Computer Vision | Human Motion Analysis | Time-Series Deep Learning | Real-Time Anomaly Detection

- Developed an end-to-end real-time fall-detection pipeline extracting human skeletal keypoints via MediaPipe Pose and a double-layer LSTM model for video input.
- Extracted 22 major body landmarks, converted each frame into a 66-dimensional feature vector and constructed 30-frame sliding-window sequential datasets for model training.
- Implemented PyTorch model training and OpenCV real-time inference, supporting skeleton rendering, risk classification and automatic warning output.

Factory Layout Design & Logistics Optimization Jun 2025

- Led on-site field research at a hydraulic steering gear factory, mapped production workflows and identified logistics bottlenecks.
- Applied SLP methodology and AnyLogic discrete-event simulation to compare multiple layout alternatives, improving material flow, space utilization and overall line efficiency.
- Delivered a 12,000-word design manual, simulation analysis report and standard A2/A3 layout schematics as final outputs.

C# WinForms Desktop Equipment Management System with SQL Server May 2025

- Built a role-based equipment-management desktop platform based on C#/.NET and ADO.NET, adopting OOP encapsulation & inheritance to create reusable functional classes.
- Integrated SQL Server database to realise full login authentication, asset query and CRUD operations, cutting manual asset lookup workload by approximately 70%.

- Added input validation and exception-handling modules, achieving stable performance over 200+ test cases; delivered a complete 30-page design report and full source code.

Mechanical Manufacturing Processes: CA6140 Shift Fork June 2024

- Designed the complete machining process route and modular fixture assembly in SolidWorks, independently developed two custom tooling structures to resolve machining constraints.
- Conducted GD&T geometric dimensioning and tolerancing analysis, alongside batch production cycle time and manufacturing cost estimation.
- Produced a 25-page design report, full engineering drawings and complete 3D model deliverables.

PROFESSIONAL EXPERIENCE

Continental Tires (China) Co., Ltd. May 2026 – Jul 2026

Industrial Engineering Intern

- Conducted on-site workflow observation and process mapping across production, assembly and packaging areas to identify bottlenecks, material-flow issues and improvement opportunities.
- Analysed production-system order data across different production lines, comparing Speed, GAP and output quantity for identical orders to identify line-efficiency gaps.
- Collected cycle time data at core workstations, performed line-balancing analysis and proposed targeted efficiency improvement schemes.
- Applied SMED methodology, created Gantt charts for the changeover process, and proposed improvements that reduced each changeover by approximately 20 minutes.
- Supported lean improvement initiatives, including workstation layout rearrangement, visual KPI management, 5S mouth-plate storage improvement, SOP compilation and CAD-based protective-cover design for environmental compliance.

Volkswagen Group (China) Technology Company Ltd. Oct 2025 – Jan 2026

Battery Management System Intern

- Participated in low-level BMS control-logic testing, supporting input-signal validity verification for the battery-balancing module.
- Used Simulink to reproduce and simulate basic BMS control programs, supporting model-based verification and winter extreme-cold vehicle testing.
- Maintained engineering guides, specifications and Confluence wiki pages, while supporting cross-team communication with colleagues and partner suppliers.

Siemens Energy Industrial Turbomachinery Dec 2024 – Feb 2025

Mechanical Design R&D Intern

- Utilised SolidWorks for 3D component modelling and ANSYS FEA simulation to optimise structural performance of turbine core parts.
- Analysed material properties and adjusted key process parameters to support manufacturing process iteration with cross-functional engineering teams.
- Collated experimental test data and compiled standardised technical analysis reports to support data-driven R&D decision-making.

MCC Huludao Non-Ferrous Metals Group Co., Ltd. Jul 2024 – Aug 2024

Assistant Mechanical Engineer

- Conducted preliminary data statistics and research to assist new product development projects.
- Supported senior engineers throughout equipment design, assembly, functional testing and routine maintenance inspection.
- Collated historical equipment operation logs, sorted failure data and summarised actionable maintenance optimisation suggestions.

TECHNICAL SKILLS

- Computer Vision & ML: Python, PyTorch, OpenCV, MediaPipe, LSTM, pose/keypoint-based motion analysis, real-time video inference
- CAD/CAE & Simulation: SolidWorks (Proficient), AutoCAD, ANSYS (FEA), AnyLogic, MATLAB, GD&T Geometric Dimensioning & Tolerancing
- Programming & Data: C/C++, C# WinForms, SQL Server, ADO.NET, Power BI, Advanced Excel
- Manufacturing Systems: ERP/MES concepts, PLC ladder logic, SLP facility layout, process improvement, workflow analysis, lean production line balancing
- Languages & Other: Mandarin (Native), English (IELTS 6.5), Piano Level 10

LEADERSHIP & AWARDS

Extracurricular & Volunteer Activities

- Sports Department Coordinator, Student Union: Planned and delivered 5+ annual campus events covering over 300 participants, lifting student engagement by 40%; judged aerospace model competition with 15+ entries and coached the university team to claim the 2023 inter-college championship.
- Member, University Football Team: Took part in multiple university-level tournaments to cultivate cross-team collaboration and rigorous execution awareness.
- Volunteer, Huludao Wild Bird Conservation Association: Completed biodiversity field surveys with GPS tracking and delivered public science education for wetland protection.

Awards & Honours

- Third Prize, The 2nd International English Competency Challenge for College Students
- First Prize, Class Emblem Design Competition, School of Mechanical Engineering